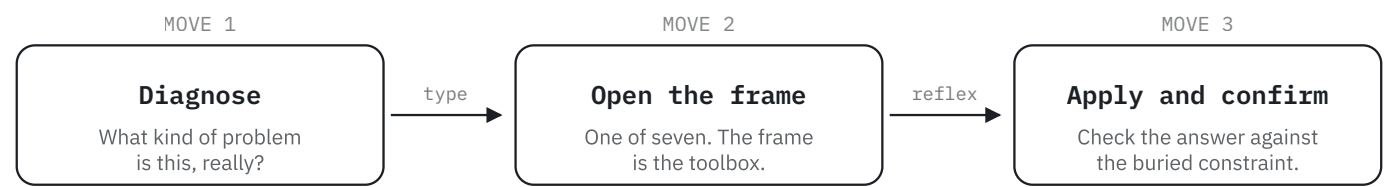


CCAR-F Frame Map

A decision framework for the Anthropic **Claude Certified Architect – Foundations** exam. Its questions drop you into a concrete situation, sometimes a system misbehaving, sometimes a design choice where no option is obviously wrong, and ask for the best decision. This map turns that into three moves: **read the situation, open the frame it belongs to, apply its reflex**. It complements the official Exam Guide, it does not replace it.

THE METHOD



Read the whole scenario before the options. The clause that decides the answer is usually buried mid-paragraph, not in the last line.

Run the moves in order. Never pick a frame before reading the case.

MOVE 1 · THE ROUTER

Classify before you medicate. The wording on the left is what the scenario tends to say; the chip on the right is where it belongs. A question can span two frames, in which case split it and treat each part separately.

inconsistent output · wrong format · invents values · can't guarantee a step · mishandles failures	1 · Output Reliability
find · trace · locate · which exploration strategy · next step to understand this code	2 · Codebase Navigation
coordinator · subagents · how information flows · decompose · parallel or sequential	3 · Orchestration
approaching the context limit · long session degrades · cached results are stale · resume or fork	4 · Context & State
where should this live · who should see it · CLAUDE.md, rules, skills, commands, settings	5 · Configuration Placement
the customer asks for a person · outside documented authority · irreversible action	6 · Human Escalation
balances cost and latency · plan or direct · sync or batch · automate or keep review	7 · Execution Mode
repeated calls just to discover <i>what exists</i> before doing anything	expose a resource, not a tool

Don't let the decorative technology intimidate you. Strip the jargon and ask what decision sits underneath: it is almost always one of the seven. The frame names here are study scaffolding, not official Anthropic taxonomy.

The Seven Frames

Each frame is a **type of decision**, never a technology. The reflexes below are defaults: a constraint buried in the scenario can overturn any of them.

1 Output Reliability

The output can't be trusted. What actually fixes it?

A tool exists and nobody calls it → **expand its description**

Two tools get confused for each other → **rename both**, update both

One tool, output shape varies → **split it** into typed tools

Descriptions are fine and it still misroutes → check the **system prompt**

must · never · before X runs → **deterministic mechanism**, not prose

Retries what can't succeed → **structured error metadata**

2 Codebase Navigation

How do I explore code I don't know yet?

Searching content → **Grep** · searching names → **Glob**

Build understanding **incrementally**: entry points, then follow imports and trace

Wrappers re-export under new names → **list every exported name first**

Edit fails on a non-unique match → **Read, modify, Write**

It doesn't fit the window → **scoped subagents** returning compact reports

3 Orchestration

Several agents. How does information flow?

Subagents never talk to each other; **everything routes through the coordinator**

Their context is **isolated** → put every constraint in their own prompt

Delegate **goals and quality criteria**, not step-by-step procedure

Parallel work → several delegations **in one response**, not across turns

Gaps after synthesis → **assess coverage, re-delegate, re-synthesize**

4 Context & State

What lives in the window, and what must survive it?

Near the limit → **manage what's in it**. Restarting with nothing carried over is the one move that always loses

Bloated tool output → **prune fields**, keep exact values

Long multi-topic conversation → **summarize what's resolved**, keep the live thread verbatim

Cached results are stale → **fresh session, seeded with a summary, then refetch**

5 Configuration Placement

Where does this instruction live, and who sees it?

Tied to a **path** → `.claude/rules/`, glob in the frontmatter

A **procedure with steps** → a skill, loaded on demand

A **universal standard** → `CLAUDE.md`, always loaded

Invoked by hand → a slash command in `.claude/commands/`

A **hard guarantee** → a hook, never a prompt

Second axis: the project directory is the team, the home directory is you alone

6 Human Escalation

When does a person take over?

They asked for a person → **honor it immediately**, no investigation first

Frustrated but hasn't asked, and it's within your capability → **acknowledge and offer to resolve**; escalate if they insist

Policy is **silent or ambiguous** on what they want → escalate

Several possible matches → **ask for an identifier**, don't guess

The handoff is a package: who, root cause, amount, recommended action

Never sentiment or self-reported confidence as the trigger

7 Execution Mode

Nothing is broken. In what mode do I run this?

Multi-file, architectural, or several valid approaches → **plan first**; a clear single-file fix → **go direct**

Blocking, or needs tools mid-request → **synchronous**

Overnight and latency-tolerant → **batch**

Who reviews the work → an **independent instance**, not the one that produced it

Three Lenses

The frame gives you the toolbox. These three questions run **on top of every frame**, and they are what stop a good reflex from being applied in the wrong world, at the wrong altitude, or in the wrong place.

THREE LENSES THAT CROSS EVERY FRAME

Build vs Use

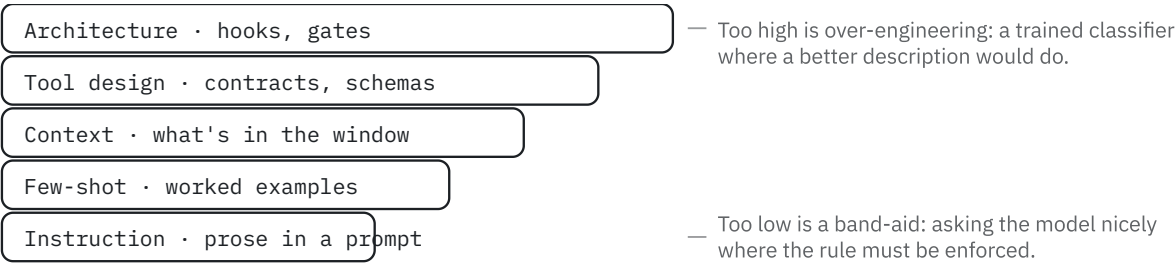
Am I **building** the agent with an SDK, where the layers live in code, or **using** a finished coding agent, where configuration lives in files? A product for end users is always the first, even when the product writes code.

Root vs Symptom

Fix it where the value is born, with the source document still in view. Downstream work is legitimate to **flag** a problem, never to quietly paper over it. Deterministic is not the same as being in the right place.

The Reliability Ladder

Climb only as far as the failure justifies. The test is not “I want this to always happen”; it is **what happens if it fails once**. Money moved wrongly buys a gate. One wasted API round trip does not.



Requests ask. Mechanisms impose.

TWO HABITS THAT DECIDE MORE QUESTIONS THAN ANY SINGLE FACT

Exoneration clauses

When a scenario tells you a component was reviewed and is sound, or that syntax errors are gone, it is not reassurance. It is telling you **where the cause is not**, and it usually eliminates two options before you have read them.

Multi-part cases

Count the problems before you look at the options, then map one fix to each. If both of your picks aim at the same half, **one of them is wrong by construction**, however good it sounds.

Discriminators

The frame narrows the answer to two or three defensible options. These are what separate them. **Filled boxes are the move, dashed boxes are the plausible trap,** and outlined boxes are the steps of a procedure you run in order.

THE PAIRS THAT DECIDE THE HARD QUESTIONS

Values keep coming back inconsistent. Constrain the field, or split the tool?

INPUT IS OPEN

A field or parameter accepting anything → **enumerate the allowed values**, and add an escape value for genuinely novel and genuinely unclear cases.

OUTPUT SHAPE VARIES

One tool returning a table, then an object, then prose → **split it** into tools with declared return shapes.

TRAP

Constraining the input when what varies is the output. The consumer keeps breaking.

The agent keeps burning calls. Better descriptions, a resource, or narrower access?

PICKS THE WRONG TOOL

Among tools it is entitled to use → **a fuller description**: inputs, outputs, and when to use this one over the neighbor.

ASKS WHAT EXISTS

Repeated exploratory calls to list what's available → **expose the catalog as a resource**.

USES WHAT ISN'T ITS JOB

Crossing a role boundary → **scope its tool set**, keeping the frequent narrow need.

TRAP

Caching. It makes the same calls faster; a resource makes them unnecessary.

Judgment is inconsistent between runs. Show it, write it, or have it check itself?

ENUMERABLE

The team can write the conditions and each is checkable → **explicit criteria**, checked one at a time.

DEMONSTRABLE

Granularity, where to cut, ambiguous phrasing → **a few worked examples** that show why one reading beat the other.

NEITHER

Gaps vary unpredictably and no stable list exists → **have it critique its own draft**.

TRAP

"Be more conservative." An intensity dial is not a criterion, and it silences good findings too.

Asynchronous batch, or synchronous? Run the filters in order and stop at the first hit.

FILTER 1

Is anything blocked waiting on the result? **Synchronous**.

FILTER 2

Does the available window cover the full processing window? If not, **synchronous**.

FILTER 3

Does it need tool calls mid-request? Batch **cannot** do that.

ONLY THEN

Nothing fired → **batch**.

TRAP

Reaching for the 50% saving before running the filters. The saving is what you gain, never what decides.

Extraction failed validation. Will retrying with feedback help?

YES, IT'S SHAPE

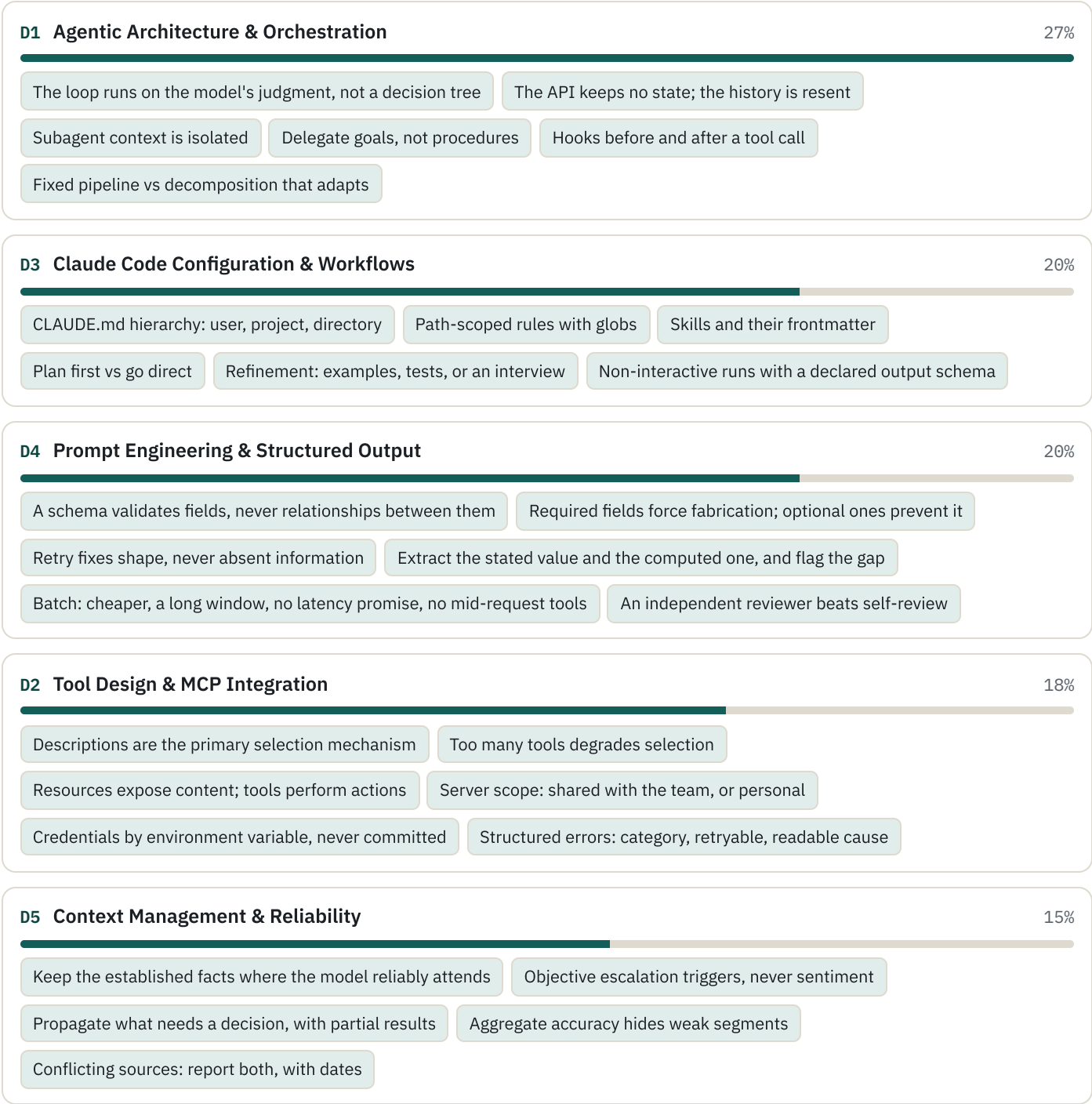
Wrong format or structure → **retry with the failed output and the specific errors attached**.

NO, IT'S ABSENT

The value only exists in a document you never supplied. Retrying pushes the model to invent it. Make the field optional and let it return nothing.

The Five Domains

Weights are from the official blueprint. Four of the six scenarios are drawn per sitting. **Study by weight, not by comfort.**



What this is. An independent study aid written while preparing for the exam. It is not affiliated with, endorsed by, or reviewed by Anthropic, and it reproduces no exam content. The frame and lens names are study scaffolding invented for this map, not official terminology.

How to use it. Read the official Exam Guide first and treat this as the layer on top: the guide tells you what is examinable, this tells you how to decide once you are looking at four defensible options. Print to PDF from your browser for a five-page handout.